

Credit Card Default Auto-ML Report — lgbm (sub)

🕒 2026-05-28 22:52:41 UTC

모형 lgbm

roc_auc = 0.7791

BEST(catboost) = 0.7796

Δ vs best = -0.0005

1 점수 요약 (본 모형 vs best)

metric	lgbm (현재)		catboost (BEST)		Δ(Test)
	Train	Test	Train	Test	
roc_auc	0.8144	0.7791	0.8099	0.7796	-0.0005
ks	0.4739	0.4349	0.4660	0.4362	-0.0013
lift	3.2775	3.1123	3.4037	3.1274	-0.0151
accuracy	0.8252	0.8168	0.8308	0.8177	-0.0008
precision	0.7053	0.6652	0.7133	0.6589	+0.0063
recall	0.3601	0.3459	0.3931	0.3640	-0.0181
f1	0.4768	0.4551	0.5069	0.4689	-0.0138
pr_auc	0.6021	0.5563	0.6194	0.5588	-0.0025

2 Feature Importance — lgbm

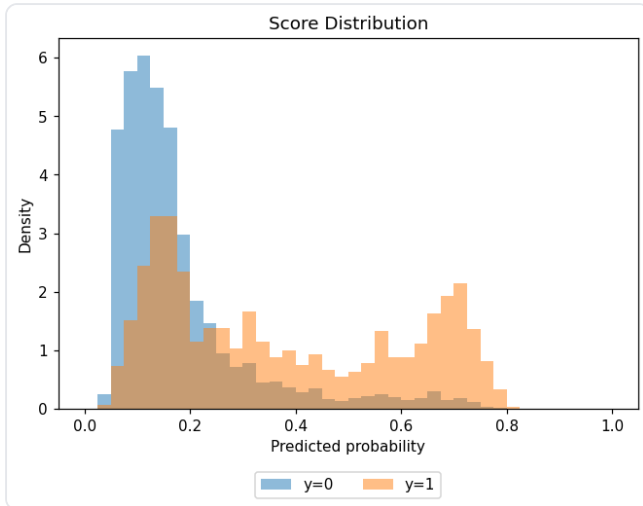
relative · top 19 · 전체 분포는 feature_importance.csv

Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	0.3634	16	BILL_AMT5	0.0137
2	PAY_2	0.1993	17	BILL_AMT6	0.0111
3	PAY_3	0.0564	18	PAY_AMT5	0.0109
4	LIMIT_BAL	0.0484	19	SEX	0.0027
5	BILL_AMT1	0.0424			
6	PAY_5	0.0362			
7	PAY_AMT2	0.0350			
8	PAY_AMT3	0.0324			
9	PAY_AMT1	0.0285			
10	BILL_AMT2	0.0270			
11	PAY_4	0.0222			
12	PAY_6	0.0192			
13	PAY_AMT4	0.0184			
14	AGE	0.0177			
15	PAY_AMT6	0.0151			

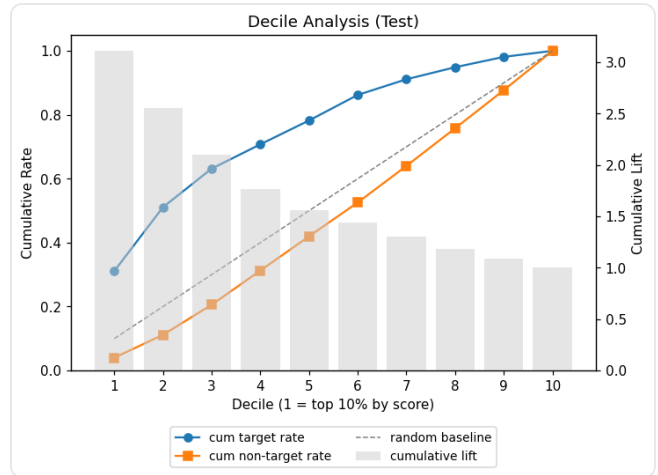
3 모델 상세 — lgbm

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.547 ~ 0.801	600	413	187	413	187	600	10.0%
2	0.314 ~ 0.547	600	265	335	678	522	1200	20.0%
3	0.223 ~ 0.314	600	160	440	838	962	1800	30.0%
4	0.178 ~ 0.223	600	101	499	939	1461	2400	40.0%
5	0.155 ~ 0.178	600	99	501	1038	1962	3000	50.0%
6	0.135 ~ 0.155	600	106	494	1144	2456	3600	60.0%
7	0.114 ~ 0.135	600	65	535	1209	2991	4200	70.0%
8	0.095 ~ 0.114	600	50	550	1259	3541	4800	80.0%
9	0.075 ~ 0.095	600	43	557	1302	4098	5400	90.0%
10	0.046 ~ 0.075	600	25	575	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4442	231
True 1	868	459

4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature_selection.csv)

Feature		Selection Frequency	채택
LIMIT_BAL		100.0%	✓ 채택
AGE		100.0%	✓ 채택
PAY_0		100.0%	✓ 채택
PAY_2		100.0%	✓ 채택
PAY_3		100.0%	✓ 채택
PAY_4		100.0%	✓ 채택
BILL_AMT1		100.0%	✓ 채택
PAY_AMT1		100.0%	✓ 채택
PAY_AMT2		100.0%	✓ 채택
PAY_AMT3		100.0%	✓ 채택
PAY_AMT6		100.0%	✓ 채택
PAY_5		98.8%	✓ 채택
BILL_AMT5		98.8%	✓ 채택
PAY_6		97.5%	✓ 채택
PAY_AMT4		96.2%	✓ 채택
PAY_AMT5		95.0%	✓ 채택
SEX		91.2%	✓ 채택
BILL_AMT6		88.8%	✓ 채택
BILL_AMT2		71.2%	✓ 채택
BILL_AMT3		56.2%	제외
BILL_AMT4		28.7%	제외
EDUCATION		21.2%	제외
MARRIAGE		15.0%	제외

5 학습 설정 + 튜닝 — lgbm

키	값
tuned	YES n_trials = 15 , best_value = 0.7850
best_iteration (avg over folds)	175
params (최종)	<pre>{'objective': 'binary', 'metric': 'auc', 'verbose': -1, 'n_estimators': 800, 'learning_rate': 0.028180680291847244, 'num_leaves': 26, 'max_depth': 9, 'min_data_in_leaf': 50, 'feature_fraction': 0.6488152939379115, 'bagging_fraction': 0.798070764044508, 'reg_alpha': 2.039373116525212e-08, 'reg_lambda': 1.527156759251193, 'min_gain_to_split': 0.12938999080000846, 'actual_max_depth': 10}</pre>